MAPS FROM THE CECH CHAIN COMPLEX OF A FAMILY INTO A PRESHEAF ARE THE CECH COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The presheaf Z_X represented by an object X of the category. Notation of this chapter: for an object X the presheaf Z_X assigns to Y the free abelian group on Hom(Y, X); the chain complex Z_{U, bullet} is built from these. Supplies the meaning of the summands of the displayed chain complex and of the alternating sign maps between them; the identification of maps out of the complex is the counted content.
- Hom out of a direct sum of representable presheaves into a presheaf. Modules on Sites, Lemma labelled sites-modules-lemma-obvious-adjointness (canonical Stacks tag 03A8); invoked at proof line 1021. Supplies the two displayed equalities Hom(sum, F) = prod Hom(representable, F) = prod F(...); the resulting term-by-term identification with the Cech complex is inside the counted span.

Source reference. Original lemma 03A8: sources/stacks-chow/tags.txt, source lines 4299–4299. Cross-chapter standard prerequisite invoked once at proof line 1021 for the adjunction between direct sums and products used to expand a map out of a sum of representable presheaves; the two displayed equalities and the comparison with the Cech complex are retained verbatim in the counted span.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be a family of morphisms with fixed target such that all fibre products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ exist in $\mathcal{C}$. Consider the chain complex $\mathbf{Z}_{\mathcal{U},\bullet}$ of abelian presheaves*

$$\ldots \to \bigoplus_{i_0 i_1 i_2} \mathbf{Z}_{U_{i_0} \times_U U_{i_1} \times_U U_{i_2}} \to \bigoplus_{i_0 i_1} \mathbf{Z}_{U_{i_0} \times_U U_{i_1}} \to \bigoplus_{i_0} \mathbf{Z}_{U_{i_0}} \to 0 \to \ldots$$

where the last nonzero term is placed in degree 0 and where the map

$$\mathbf{Z}_{U_{i_0} \times_U \ldots \times_u U_{i_{p+1}}} \longrightarrow \mathbf{Z}_{U_{i_0} \times_U \ldots \widehat{U_{i_j}} \ldots \times_U U_{i_{p+1}}}$$

is given by $(-1)^j$ times the canonical map. Then there is an isomorphism

$$\mathrm{Hom}_{PAb(\mathcal{C})}(\mathbf{Z}_{\mathcal{U},\bullet}, \mathcal{F}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$$

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functorial in $\mathcal{F} \in \mathrm{Ob}(PAb(\mathcal{C}))$.

Proof. This is a tautology based on the fact that

$$\begin{aligned}\mathrm{Hom}_{PAb(\mathcal{C})}(\bigoplus_{i_0 \ldots i_p} \mathbf{Z}_{U_{i_0} \times_U \ldots \times_U U_{i_p}}, \mathcal{F}) &= \prod_{i_0 \ldots i_p} \mathrm{Hom}_{PAb(\mathcal{C})}(\mathbf{Z}_{U_{i_0} \times_U \ldots \times_U U_{i_p}}, \mathcal{F}) \\ &= \prod_{i_0 \ldots i_p} \mathcal{F}(U_{i_0} \times_U \ldots \times_U U_{i_p})\end{aligned}$$

see Modules on Sites, Lemma 03A8. □